



PRIFYSGOL
BANGOR
UNIVERSITY

Module Code:	ABJ-4061	Assessment:	Coursework: Project on Data Analysis	Weighting:	50%
Module Title:	Data Science				
Academic Year:	2023/24				
Module Coordinator:	Heather He				
Method of Submission:	Electronic via Blackboard				
Submission Due Date:	15 May 2024 23:59				

Reminder of Academic Misconduct:

You are reminded not to copy material from any sources without properly referencing it, as this constitutes plagiarism: cases of plagiarism will be referred to the Business School Academic Integrity Officer and will be subject to a deduction of marks, which can result in an overall mark of 0%.

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Work submitted after the deadline:

Unauthorised late work: Work submitted up to 7 days after the original deadline without extension will be graded and given feedback, but the grade will be capped at 50% for postgraduate students. Work submitted after 7 days without a valid extension will still be given feedback but will be awarded a 0 grade (F4).

Authorised late work: If you feel you have legitimate grounds for an extension, then you must use the Request Centre to request an extension, and you must do so by the hand-in deadline.

This assessment relates to the following module learning outcomes (LOs):

LO1: Demonstrate proficiency in the application of statistical learning techniques using the R programming language.

LO3: Implement data-driven statistical models to address big data focused business problems.

LO4: Execute data visualisation techniques to communicate solutions to management.

LO5: Produce and present technical solutions to non-technical stakeholders in a professional manner using concise written reports.

Coursework Brief

Project comprises 50% of the assessment weighting for the 20-credit module. The other element of the module assessment is an exam (50%). Please direct all your enquiries on the project to Heather He. This coursework consists of five tasks. You must complete ALL of them. This assignment is about data analysis using R and Power BI. Therefore, all data import/output, manipulation, exploration, analysis, and presentation must be conducted using R & Power BI. You will lose credit if, for example, you use Python to perform any of the tasks.

For Task 1 - 4, please load the data from a MySQL Database that is stored on your own machine into R session.

For Task 5, please load the data stored in a csv format into PowerBI.

Task 1: Understanding Market Campaign Data [20%]

You are a newly employed data scientist at BangorTelco. The first question you have been asked to address is: 'Is there a way to determine in advance which customers are likely to accept the offer in the marketing campaign?'

The IT team provide you with access to the company database, which contains a table (BangorTelco_MarketCampaign) containing data on over 20,000 customers, randomly selected from a larger database of previous customers, including whether they have accepted the offer in the last campaign. The data in this table are:

ID	Customer's unique identifier
Year_Birth	Customer's birth year
Education	Customer's education level
MaritalStatus	Customer's marital status
Income	Customer's yearly household income
Recency	Number of days since customer's last purchase
NumWebPurchases	Number of purchases made through the company's website
NumStorePurchases	Number of purchases made directly in stores
NumWebVisitsMonth	Number of visits to company's website in the last month
Response	1 if customer accepted the offer in the last campaign, 0 otherwise

Your goal is to create a decision tree which can predict class membership of the Response variable. The sales team will then use your tree's rules to determine whether a customer is likely to accept the offer in the next campaign. Clearly, the better the tree prediction, the more marketing costs can be reduced.

N.B. You may encounter certain attributes that contain abnormal data points (e.g., Year_Birth entries earlier than 1900 appear to be input errors). Please handle these abnormal data points with careful consideration to ensure the reliability of your model.

Task 2: Logistic Regression [20%]

Management was impressed with your work on Decision Trees and have heard it is possible to predict the probability directly from the underlying data.

Your goal is to build the best logistic regression model you can to predict the probability a given customer will accept the offer.

Task 3: k Nearest Neighbours [20%]

Management was impressed with your work on Decision Trees and Logistic regression. They have heard it may be possible to infer the probability a person is likely to accept the offer from whether they have similar characteristics to other known customers who have already left.

Your goal is to build the best k Nearest Neighbours model you can to predict the probability a given customer will accept the offer.

Task 4: Clustering [20%]

Previously, management have been quite directed about what they were looking for and what they wanted to predict. Now, they are asking you what kind of things can be 'discovered' from the data. In particular they are interested in whether there are any kind of natural groupings that exist within their customers.

Your goal is to find and explain any ONE natural grouping you find within the data. You only need to concentrate on finding one way to group users, and then explain that grouping in business terms.

Task 5: Building a Data Science Dashboard [20%]

Management was impressed with the models you have built so far!

They are also starting to think about deployment. At a recent conference on Data Science and Big Data, management attended a seminar about Microsoft Power BI, and they think this would be the perfect deployment approach for their models.

For this task, you only need to implement the tree, logistic regression, and KNN models.

Management wants you to build a Data Science Dashboard. As a minimum, for the tree, logistic regression and KNN models, the dashboard will load the models and for each model, it will show visualizations of the model and its fitness for purpose. The dashboard would then allow a user to input the required details and receive a prediction from each of the underlying models.

Beyond this required functionality, you may enhance the dashboard as you see fit.

Instructions

Submit to Blackboard:

A report in .PDF format produced from knitr. Your report should contain your responses to Tasks 1-4. You must submit the report to the Turnitin “Report Submission” tab on Blackboard. The report must be produced from knitr; failure to deliver this will result in reduction to your marks. There are two approaches you could take to produce the PDF report:

- You could either knit your R Markdown file into a HTML file and then save it to a PDF file, or
 - You could directly knit your R Markdown file into a PDF file.
- **The Power BI dashboard (.pbix file).** The dashboard should be submitted to the “Dashboard Submission” tab on Blackboard. Failure to submit this file will result in reduction to your marks. Note that the Power BI software (free license) is available via University Digital Services - Microsoft Office 365.
- **An R Markdown file in .RMD format containing your codes.** The markdown file should be submitted to the “R Markdown Submission” tab on Blackboard. Failure to submit this file will result in reduction to your marks.
- Please name your files using your student ID number, e.g. 500123456.PDF, 500123456.pbix, and 500123456.RMD. You could include author name inside your report/dashboard/R Markdown file, but **do not include your name in your file names.**
- You are allowed to test submit your report before the due date. You can use Turnitin to check your assignment for plagiarism before you submit your final version. The last submission prior to the deadline will be treated as the final submission and will be the copy that is assessed by the marker.